



IN410

Cryptography and Secure Communications

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Credits

- The following slides are the adaptation of
 - Lecture slides of Lawrie Brown based on William Stallings book “Cryptography and Network Security: Principles and Practice”
 - Lecture Slides of Dan Boneh – Stanford University
 - Lecture slides of Pawel Wocjan – University of Central California
 - Lecture Slides of Ahmed Serhrouchni – Telecom ParisTech
 - Others (References in the note section)



Course Syllabus

- Context
- History/ Classical ciphers
- **Symmetric Encryption ←**
- Asymmetric Encryption
- Cryptographic hash functions
- Key Distribution
- Security Protocols



Symmetric Encryption

- Block Ciphers
- Stream Ciphers
- **Modes of Operation ←**



Modes of Operation



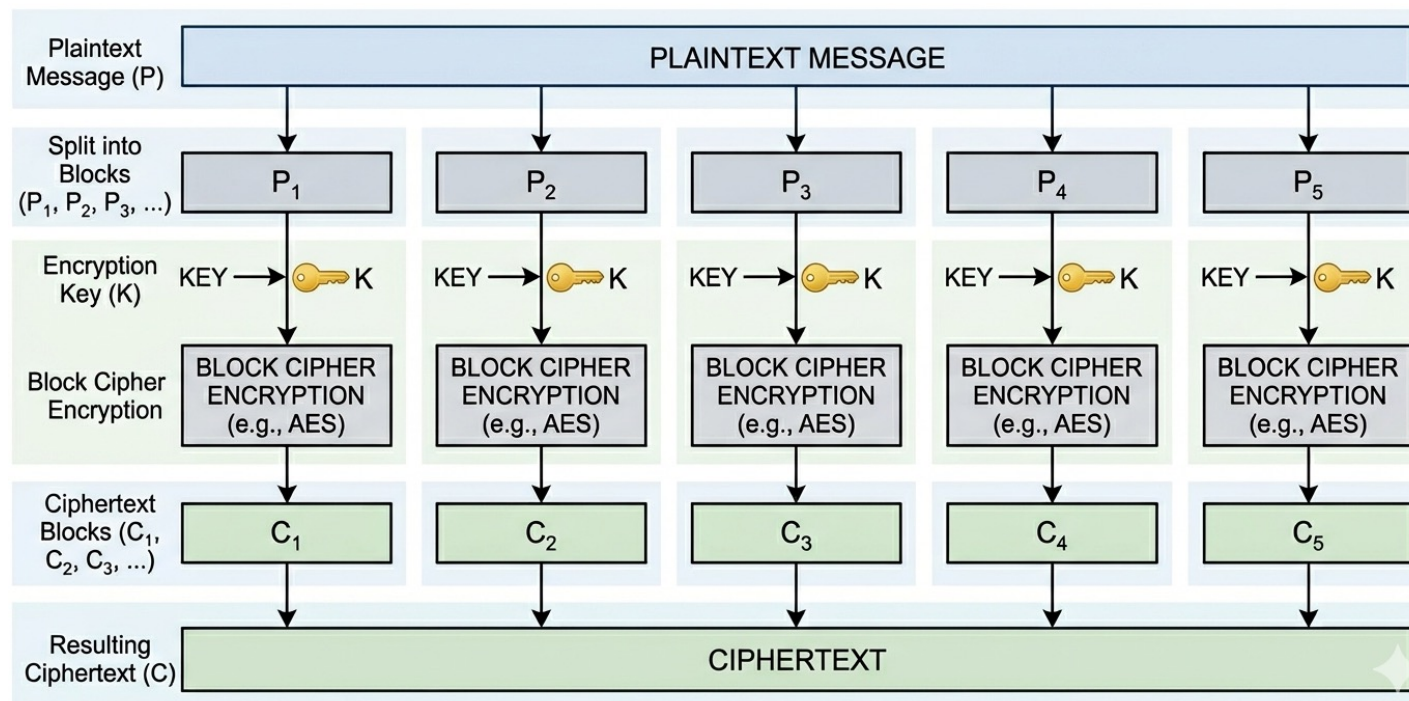
Why Do We Need Modes of Operation?

- We have studied block ciphers (e.g., AES)
- **A block cipher:**
 - Takes a fixed-size block (e.g., 128 bits)
 - Produces a fixed-size ciphertext block
- But in reality:
 - Messages are long and variable in size
- Key question:
 - How do we securely encrypt arbitrary-length data using a block cipher?



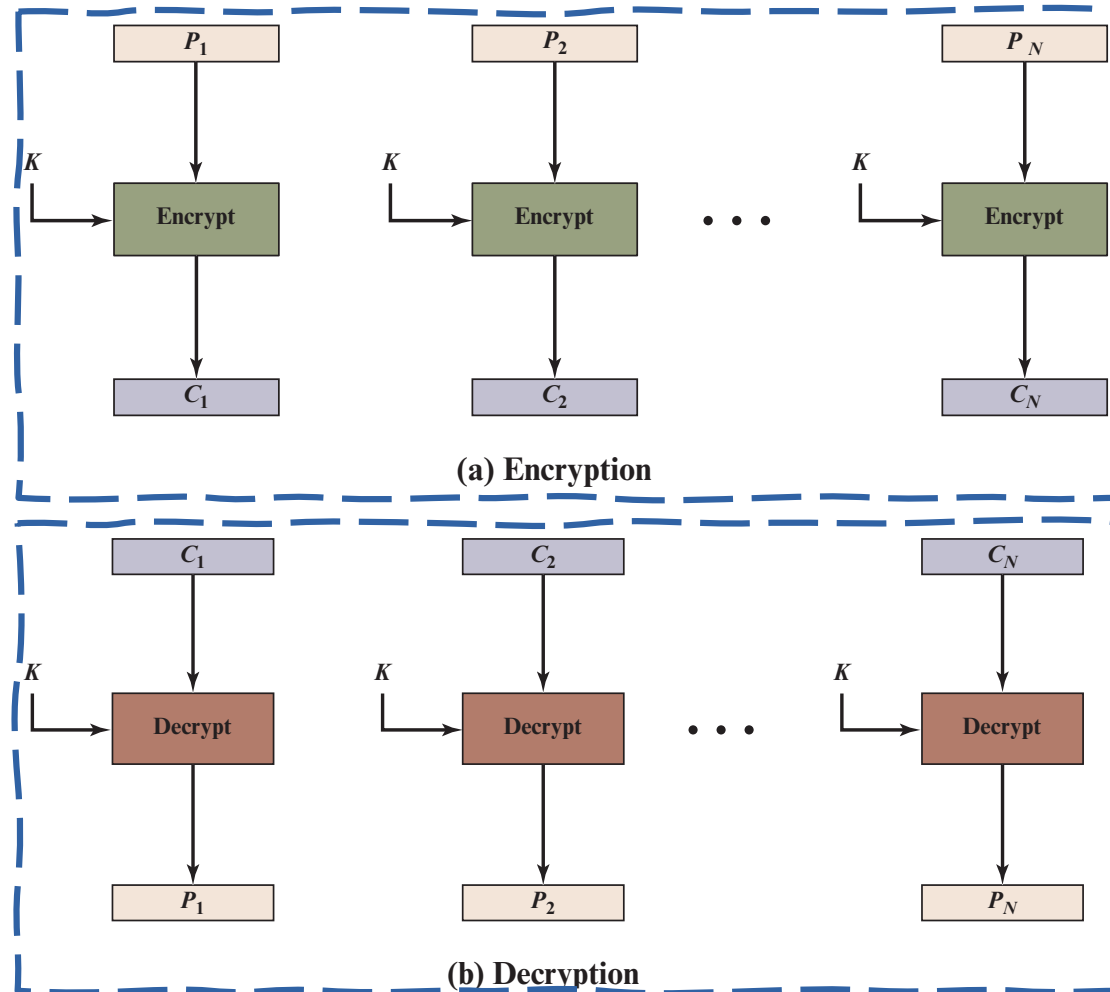
First Attempt: Encrypt Block by Block

- Message is broken into **independent blocks** which are encrypted
- Each block is encoded independently of the other blocks
- This defines the **Electronic Codebook (ECB)** approach





Electronic Codebook Book (ECB)



$$C_i = E_K(P_i)$$

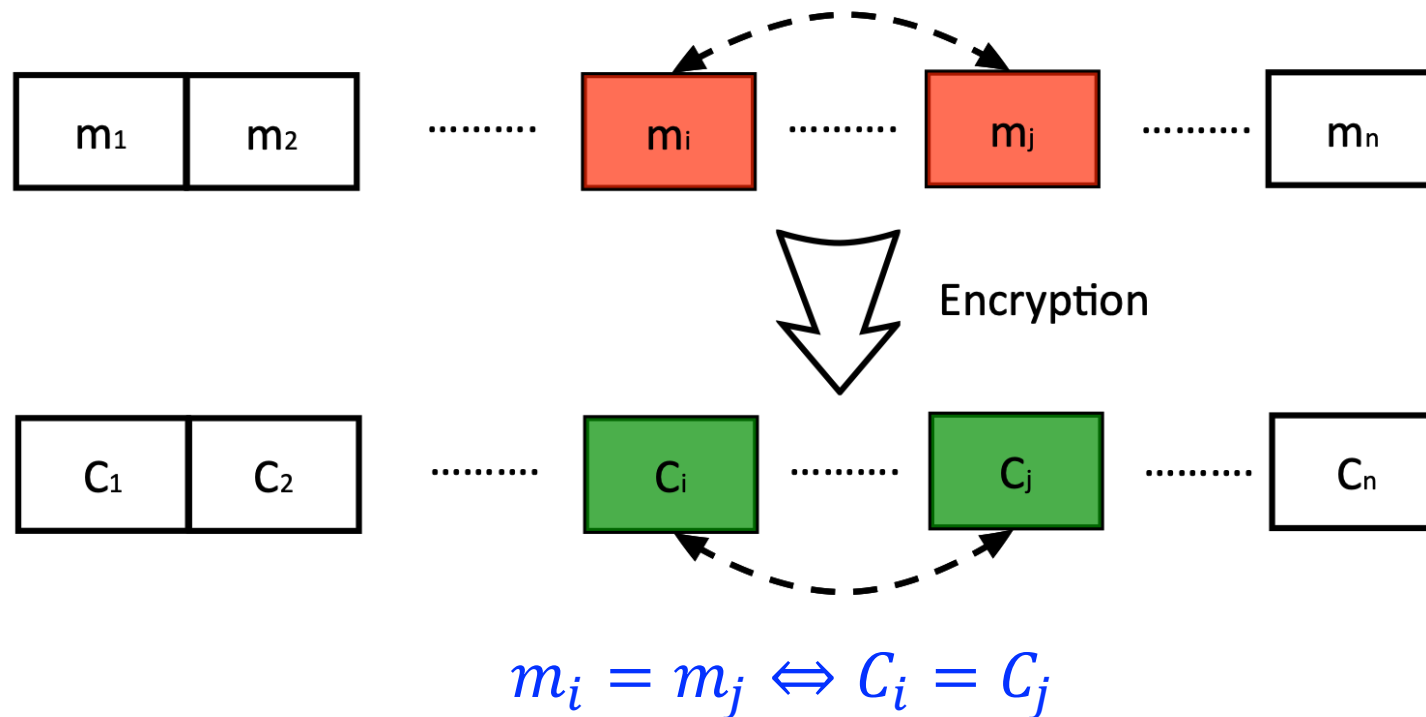
$$P_i = D_K(C_i)$$

Looks simple... but is it secure?



Why ECB Mode is Insecure?

- Identical plaintext blocks $\rightarrow$ identical ciphertext blocks



Encryption is applied... but information is still leaking



Why ECB Mode is Insecure?

- ECB is especially weak for:
 - **Highly structured messages:** Fixed formats, predefined headers
 - **Repetitive data:** Patterns with repetition aligned to block size
 - **Multimedia data** (e.g., images): Visual patterns remain visible after encryption



Original Image



**Encrypted Using
ECB mode**



**Encrypted Using
other mode**

Security Requires More Than Just Encryption



- A **secure encryption scheme** should:
 - Hide patterns
 - Prevent information leakage
- Block ciphers alone are not designed to handle long messages directly
- We need a mechanism to:
 - Chain blocks
 - Introduce randomness
 - Ensure security across the whole message



What is a Mode of Operation?

- “A block cipher mode of operation (mode for short) is a method that uses a block cipher to provide an information service, such as confidentiality or authentication.” – [NIST](#)
- It defines:
 - How blocks are processed
 - How they depend on each other
 - How randomness is introduced
- **A mode transforms a block cipher → full encryption scheme**

Criteria for Evaluating Modes of Operation



Pattern Resistance

- Does the mode prevent identical plaintext blocks from producing identical ciphertext?
- Does it hide structure and repetition in the data?

Inter-Block Dependency

- Are blocks processed independently?
- Or does each block depend on previous blocks?

Parallelizability

- Can multiple blocks be processed simultaneously?
- Or must encryption/decryption proceed sequentially?

Error Propagation

- How does an error in one ciphertext block affect decryption?
- Is the impact limited or does it spread to other blocks?

NIST Approved Operation Modes



SP 800-38A: Five Confidentiality Modes

- ECB (Electronic Code Book) ✓
- CBC (Cipher Block Chaining)
- CFB (Cipher Feedback Block)
- OFB (Output Feedback Block)
- CTR (Counter)

SP 800-38B: An Authentication Mode

SP 800-38C: An Authenticated Encryption Mode

SP 800-38D: A High-Throughput Authenticated Encryption Mode

SP 800-38E: A Confidentiality Mode Designed for Storage Devices

SP 800-38F: Methods for Key Wrapping

SP 800-38G: Methods for Format-Preserving Encryption



From ECB to CBC

- ECB encrypts blocks independently
- Identical plaintext $\rightarrow$ identical ciphertext
- Patterns are preserved $\rightarrow$ insecure

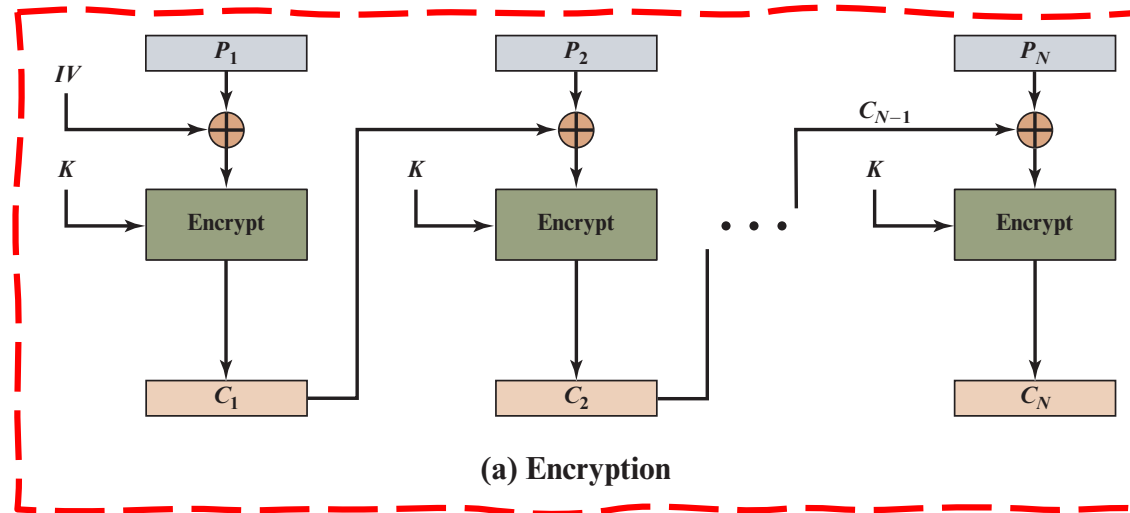
Can we make each block depend on previous ones?



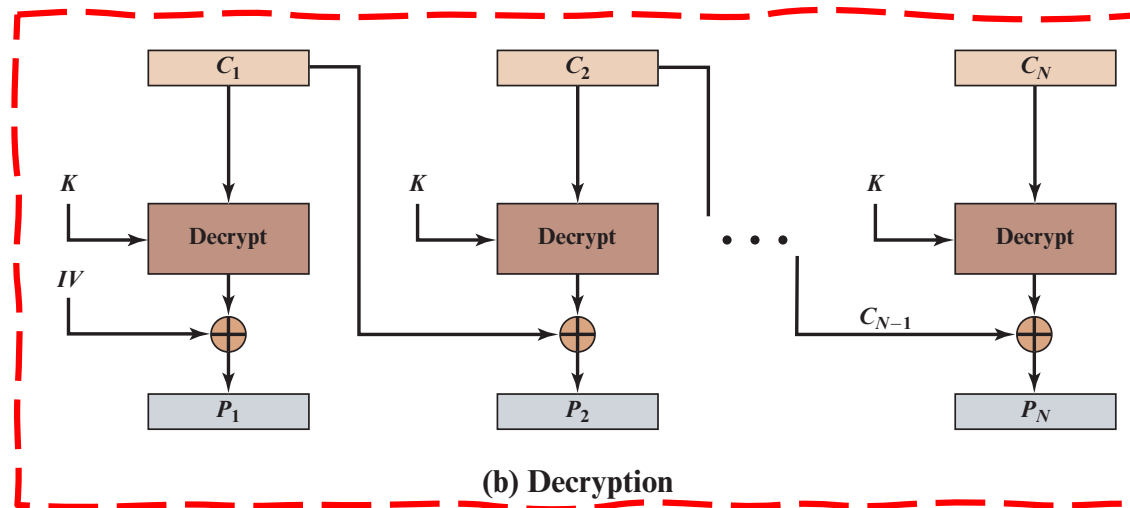
Cipher Block Chaining (CBC)

- Message is broken into blocks
- Linked together in encryption operation
- Each **previous cipher blocks** is **chained** with current **plaintext block**
- Use **Initial Vector (IV)** to start process
 - $C_i = E_K(P_i \oplus C_{i-1})$
 - $C_{-1} = IV$

CBC – Encryption and Decryption



$$C_i = E_K(P_i \oplus C_{i-1})$$



$$P_i = D_K(C_i) \oplus C_{i-1}$$



Evaluation of CBC Operation Mode

Pattern Resistance

- Hides repetition ✓

Inter-Block Dependency

- Strong dependency (chaining) ✓

Parallelizability

- Encryption: sequential ✗
- Decryption: parallelizable ✓

Error Propagation

- Error in C_i affects:
 - P_i (fully corrupted)
 - P_{i+1} (partially affected)



CBC - Initialization Vector (IV)

Role

- First block uses IV instead of previous ciphertext
- IV ensures different ciphertext for same message

Requirements

- Size equal to the block size
- Must be known to sender & receiver
- Should be unpredictable / random
- For maximum security, IV should be protected against unauthorized changes
- IV reused → Security Weakened



Stream Modes of Operation

- **Block modes** encrypt entire block
- May need to operate on smaller units
 - Real time data
- Convert block cipher into stream cipher
 - Cipher Feedback (CFB) mode
 - Output Feedback (OFB) mode
 - Counter (CTR) mode

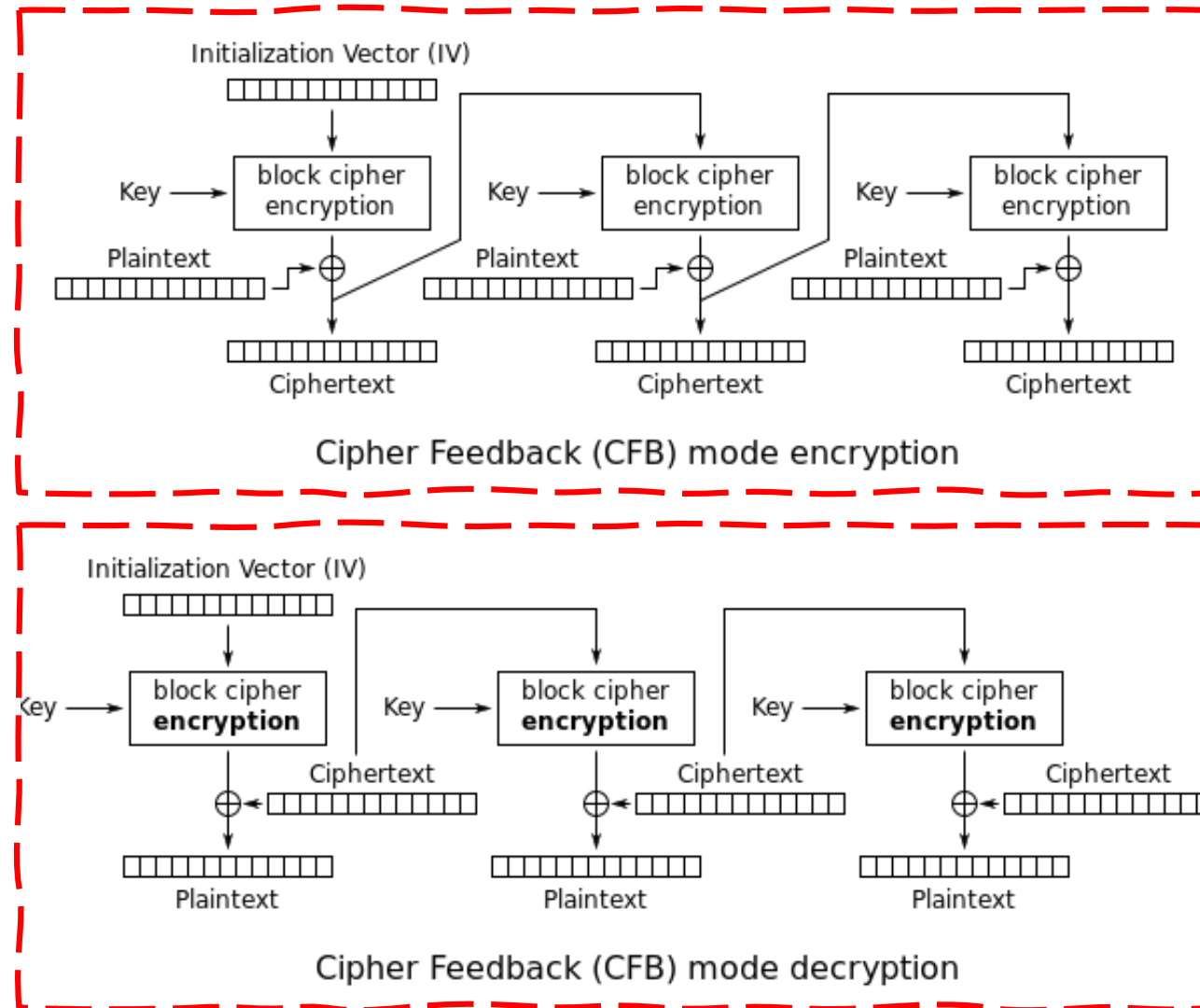


Cipher FeedBack (CFB)

- Message is treated as a stream of bits
- Added to the output of the block cipher
- Result is fed back for next stage (hence name)
- Standard allows any number of bit (1,8, 64 or 128) to be fed back (CFB-1, CFB-8, CFB-64, CFB-128)
- Most efficient to use all bits in block (64 or 128)
 - $C_i = P_i \oplus E_K(C_{i-1})$
 - $C_{-1} = IV$



Cipher FeedBack (CFB)





Evaluation of CFB

Pattern Resistance

- Hides repetition ✓
- Each block depends on previous ciphertext

Inter-Block Dependency

- Strong dependency (chaining via ciphertext) ✓

Parallelizability

- Not parallelizable ✗
- Sequential processing required

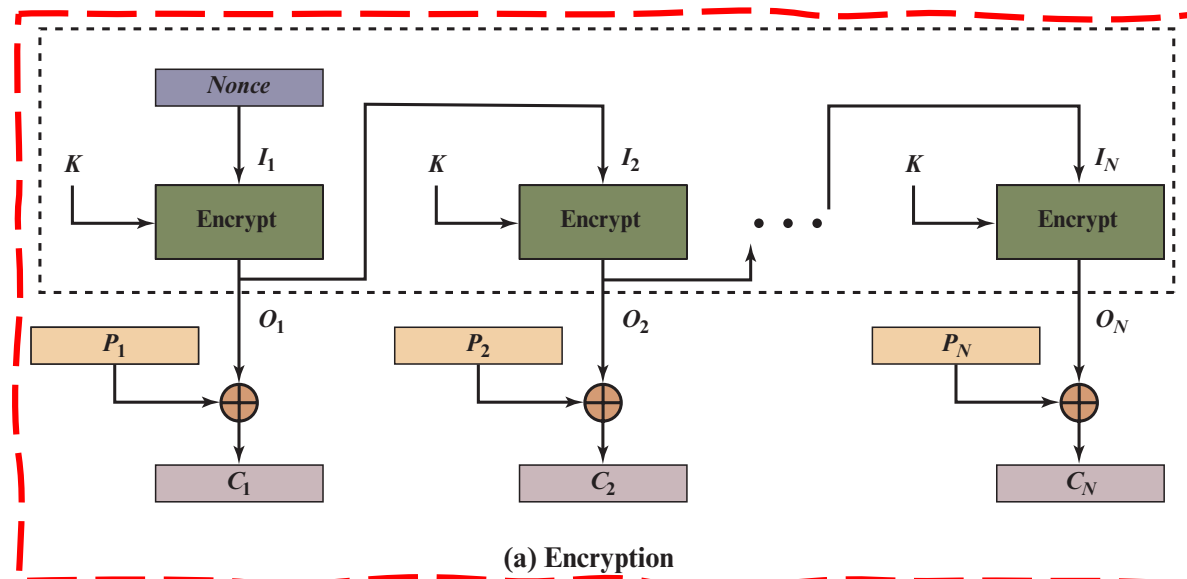
Error Propagation

- Error in C_i affects: current block P_i and next block P_{i+1}



Output FeedBack (OFB)

- Message is treated as a stream of bits
- Output of cipher is added to message
- Output is then fed back (hence name)
- Feedback is independent of message
- Can be computed in advance

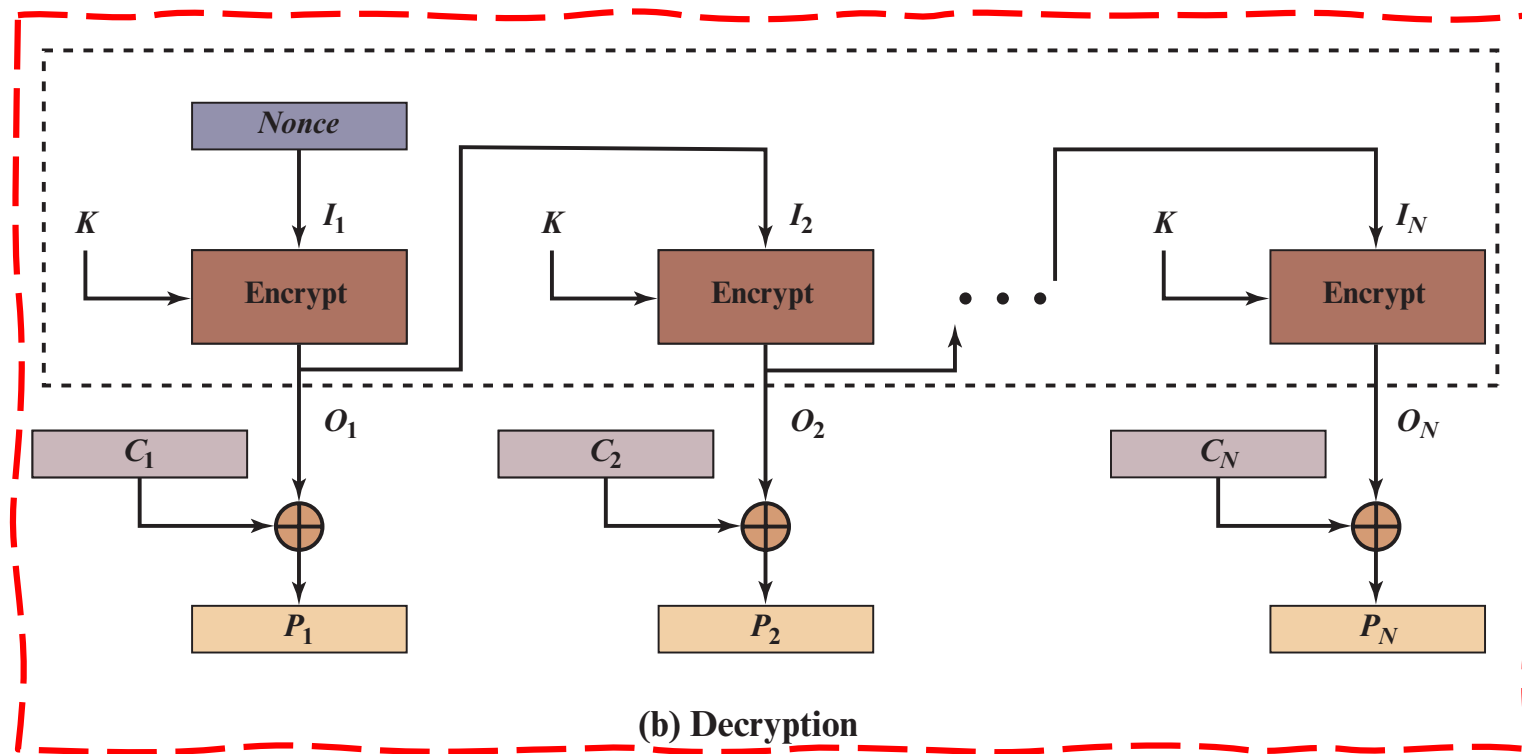


$$O_{-1} = IV$$

$$O_i = E_K(O_{i-1})$$

$$C_i = P_i \oplus O_i$$

Output FeedBack (OFB) - Decryption



$$O_i = E_K(O_{i-1})$$

$$C_i = P_i \oplus O_i$$

$$P_i = C_i \oplus O_i$$



Evaluation of OFB

Pattern Resistance

- Hides repetition ✓
- Keystream independent of plaintext

Inter-Block Dependency

- Dependency on previous internal state (keystream) ✓

Parallelizability

- Not parallelizable ✗
- keystream is sequential

Error Propagation

- Error affects only corresponding block
- No propagation to other blocks ✓



Limitations of OFB

- **IV MUST be unique for each use** (nonce)
 - If ever reused, attacker can recover outputs
- **Bit errors do not propagate**
- More vulnerable to **message stream modification**
 - Complementing a bit in the ciphertext complements the corresponding bit in the recovered plaintext.
 - **Controlled changes** to the recovered plaintext can be made.

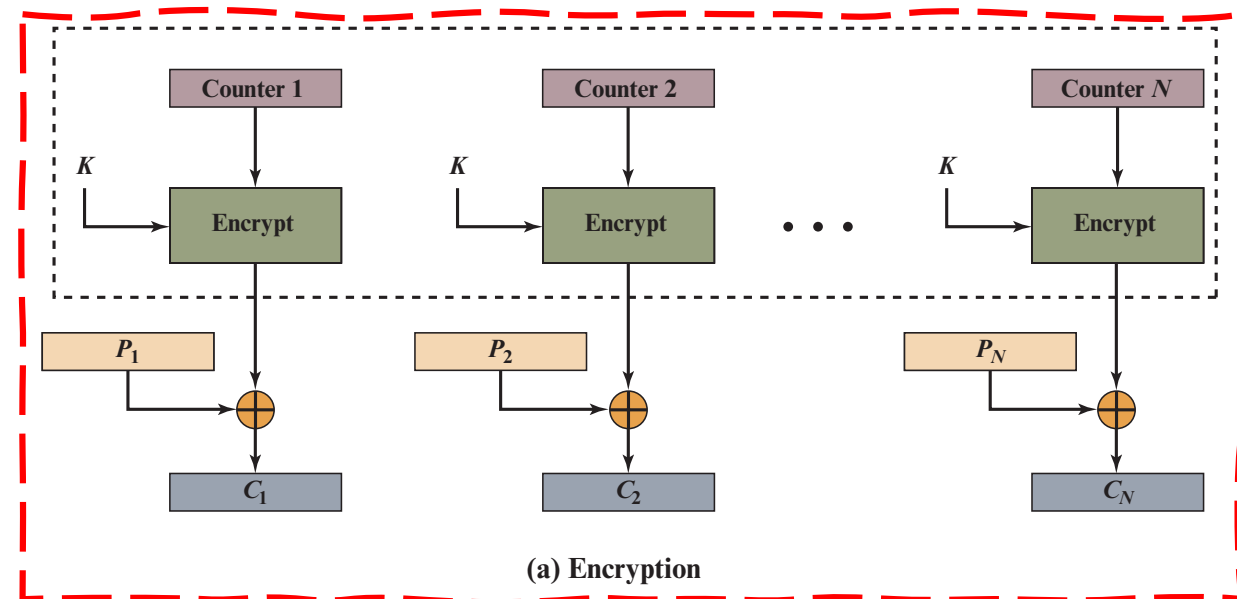


Counter (CTR)

- Similar to OFB but **encrypts counter value** rather than any feedback value
- Must have a **different key & counter value for every plaintext block** (never reused)

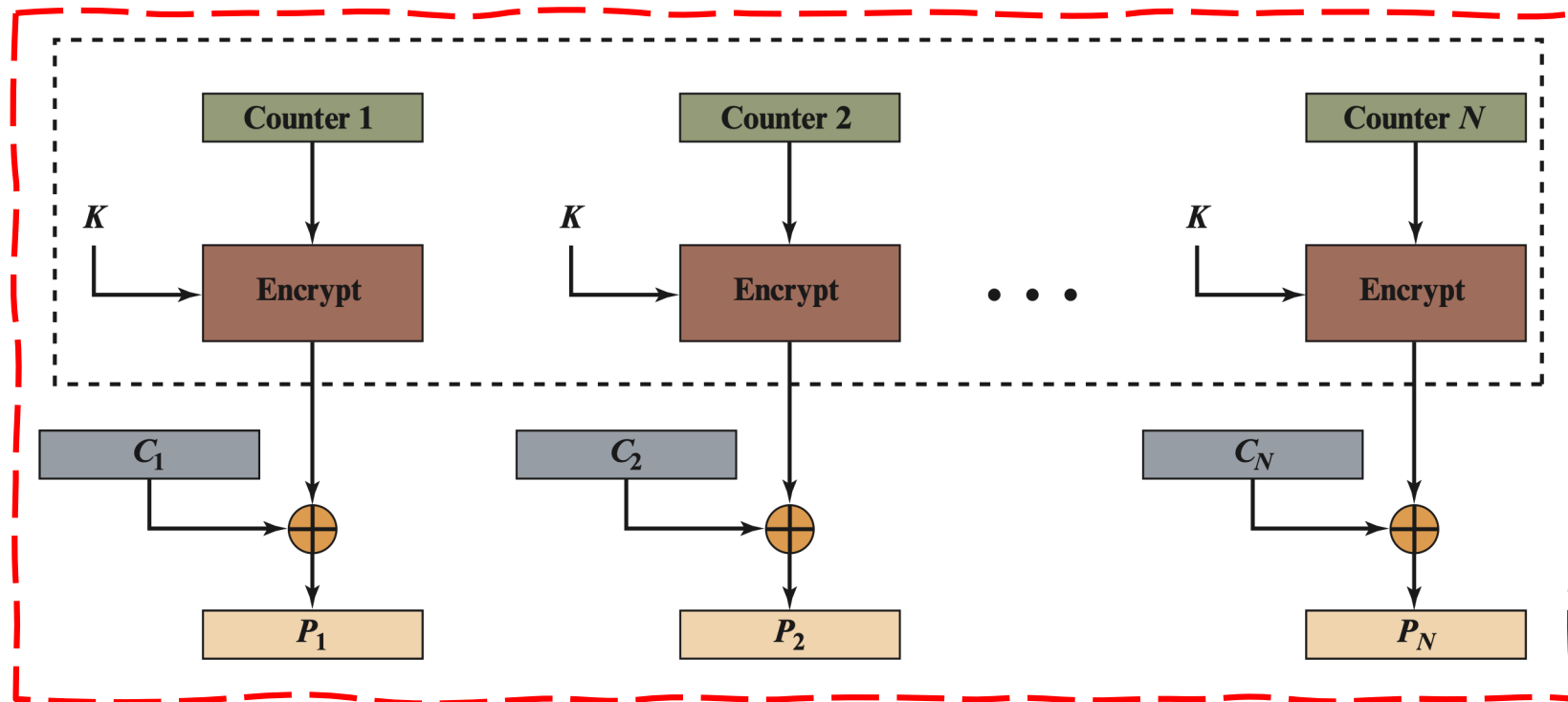
$$O_i = E_K(i)$$

$$C_i = P_i \oplus O_i$$





Counter (CTR) - Decryption



$$O_i = E_K(i)$$

$$P_i = C_i \oplus O_i$$



Evaluation of CTR Operation Mode

Pattern Resistance

- No pattern leakage (if counters are unique) ✓

Inter-Block Dependency

- No dependency (independent blocks) ✗

Parallelizability

- Encryption and decryption fully parallelizable ✓

Error Propagation

- Error affects only corresponding block ✓



Advantages of CTR

- Efficiency
 - can do **parallel encryptions**
 - can **preprocess** in advance of need
- **Random access** to encrypted data blocks
- **Provable security** (good as other modes)



Limitations of CTR Operation Mode

- **MUST ensure NEVER reuse key/counter values**
 - Same counter used → Information Leakage and Possible Plaintext Recovery
- No integrity protection
 - Vulnerable to: Bit-flipping attacks



Padding



Block Size Constraint

- Block ciphers operate on fixed-size blocks (e.g., 128 bits)
- Real messages have arbitrary length
- Problem:
 - What if the last block is incomplete?
 - Example: AES, Message size 200 bits
 - Block one : 128 bits
 - Remaining 72 bits < block size



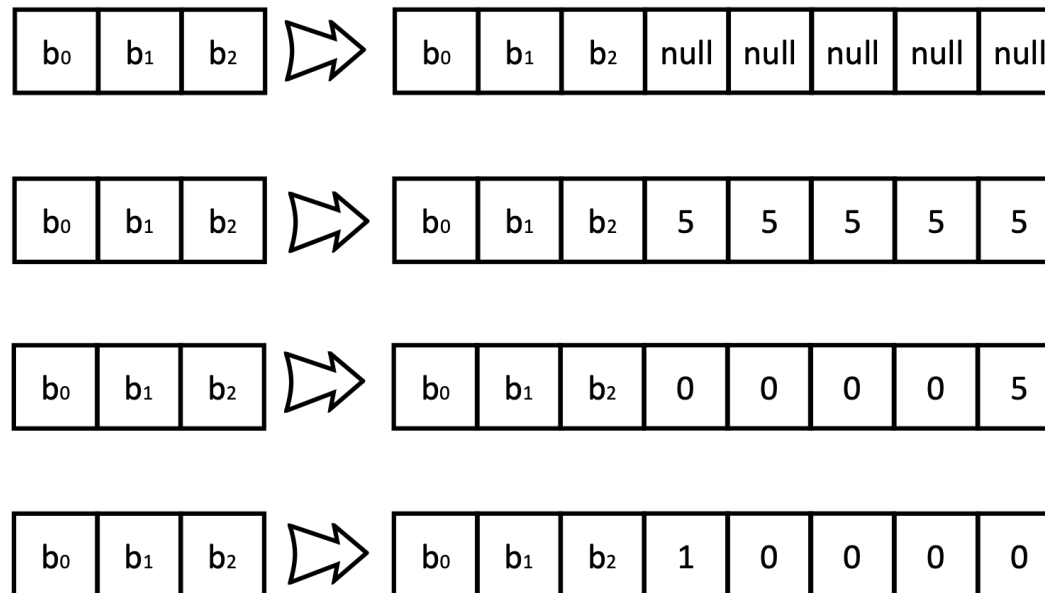
Padding Concept

- Padding adds extra bytes to the last block
- Ensures:
 - Padded Message length = multiple of block size
- Example: AES, Message size 200 bits
 - Block size: 16 Bytes, Message size: 25 Bytes
 - Pad to 32 bytes (7 Bytes padding)
- Padding is a formatting mechanism, not a security feature



Message Padding Examples

Block Size: 8 Bytes



- Some padding schemes require an extra entire block (if no pad is needed)
- There are other, more advanced modes, which avoid the need for an extra block